AI CAREERS FOR

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Online Inventory And Order Fulfillment Analysis

Project Domain

[Excel / Power BI]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

Inventory management and order fulfillment are critical components of modern e-commerce and supply chain operations. Businesses must manage inventory efficiently while ensuring that customer orders are processed and delivered accurately and on time.

This project focuses on **analyzing online inventory data and order fulfillment performance using Microsoft Excel and Power BI**. The dataset includes information about product stock levels, customer orders, shipment status, delivery time, and warehouse operations.

The project involves several stages including **data collection, data cleaning, analysis, and visualization**. Microsoft Excel is used for data preprocessing tasks such as removing duplicates, handling missing values, and performing initial analysis using formulas and pivot tables. Power BI is used to build interactive dashboards and visualize key metrics such as inventory levels, order processing time, delivery performance, and product demand trends.

The final output is an **interactive Power BI dashboard** that helps organizations monitor inventory levels, identify slow-moving products, track order fulfillment performance, and make data-driven decisions. The results demonstrate how data analytics tools can improve supply chain efficiency and customer satisfaction.

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CHAPTER 1

Introduction

The rapid growth of e-commerce and digital marketplaces has increased the importance of effective inventory management and order fulfillment systems. Businesses must ensure that products are available when customers place orders and that the orders are delivered accurately and on time.

Inventory management refers to the process of tracking, controlling, and managing stock levels in warehouses or distribution centers. The goal is to maintain the right quantity of products to meet demand while avoiding overstocking or stockouts.

Order fulfillment refers to the complete process of receiving, processing, packing, shipping, and delivering customer orders. Efficient fulfillment ensures that customers receive their products quickly and accurately, which improves customer satisfaction.

Modern businesses increasingly rely on **data analytics tools such as Excel and Power BI** to analyze inventory performance and optimize order fulfillment processes.

1.1 Problem Statement

Many businesses struggle with managing large volumes of inventory and order data. Common problems include:

- Inaccurate inventory records
- Delays in order processing
- Stock shortages or overstocking
- Lack of real-time visibility into warehouse operations

Without proper data analysis, companies cannot identify inefficiencies in their inventory management or order fulfillment processes.

This project aims to address these challenges by analyzing inventory and order data using Excel and Power BI.

1.2 Motivation

The motivation for selecting this project comes from the growing importance of **data-driven decision making in supply chain management**.

Efficient inventory and order management systems can help businesses:

- Reduce operational costs
- Improve delivery performance
- Increase customer satisfaction

- Optimize stock levels

Learning tools such as Excel and Power BI allows students to develop practical skills in business analytics

1.3 Objectives

The main objectives of this project are:

1. To analyze inventory and order fulfillment data.
2. To clean and preprocess raw data using Microsoft Excel.
3. To identify key inventory and order performance indicators.
4. To build interactive dashboards using Power BI.
5. To provide insights that help improve supply chain efficiency.

1.4 Scope of the Project

The scope of this project includes:

- Analysis of online inventory and order fulfillment data.
- Data cleaning and transformation using Excel.
- Visualization and dashboard creation using Power BI.
- Identification of important supply chain metrics.

The project does not include advanced machine learning models or real-time inventory tracking systems.

CHAPTER 2

Literature Survey

A literature survey reviews existing research and tools used in inventory management and order fulfillment analysis..Inventory management is a critical process that involves planning, tracking, and controlling stock levels to balance supply and demand effectively.

Order fulfillment acts as the bridge between customer demand and product delivery. The process includes picking, packing, shipping, and confirming delivery.

With the growth of digital commerce, businesses increasingly use analytics tools to monitor inventory and fulfillment performance.

2.1. Tools and Technologies Review

Microsoft Excel

Excel is widely used for:

- Data cleaning
- Data transformation
- Pivot tables
- Basic visualization

Power BI

Power BI is a business intelligence tool used to create interactive dashboards and analyze large datasets.

Power BI dashboards allow organizations to monitor inventory levels, track order fulfillment performance, and identify operational inefficiencies.

2.2 Gaps Identified

Despite the availability of advanced analytics systems, many small and medium businesses still rely on spreadsheets and manual methods.

This project demonstrates how simple tools like Excel and Power BI can be used effectively for supply chain analytics.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The project follows a structured workflow:

Step 1: Data Collection

The dataset includes:

- Product ID
- Product name
- Inventory quantity
- Order ID
- Order date
- Delivery status
- Warehouse location

Step 2: Data Cleaning

Excel is used to:

- Remove duplicate records
- Handle missing values
- Standardize data formats

Step 3: Data Analysis

Data analysis is performed using:

- Pivot tables
- Conditional formatting
- Excel formulas

Step 4: Data Visualization

Power BI is used to create dashboards such as:

- Inventory overview dashboard
- Order fulfillment dashboard
- Sales and demand trends

3.2 Modules / Components Used

- Data Ingestion Module
- Data Cleaning Module
- Data Analysis Module
- Dashboard Visualization Module
- Reporting Module

3.3 Data Flow Diagram (DFD)

DFD Level 0

Raw Data → Data Processing System → Dashboard Output

DFD Level 1

1. Data Collection
2. Data Cleaning
3. Data Analysis
4. Dashboard Generation

3.4 Advantages

- Faster data analysis
- Better inventory tracking
- Improved decision making
- Reduced operational errors
- Clear visual insights

3.5 Requirement Specification

Hardware Requirements

Processor: Intel Core i3 or higher
RAM: 4GB minimum
Storage: 50GB free space

Software Requirements

Microsoft Excel
Power BI Desktop
Windows 10/11
Web browser

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Dataset Source: Online sample dataset (Kaggle / project dataset)

Original dataset size:
5000 rows, 12 columns

Issues found:

- Duplicate records
- Missing order status
- Incorrect date formats

Cleaning steps:

- Removed duplicates using Excel
- Replaced missing values
- Converted date fields into proper format

4.2 Visualization / Output Results

Power BI dashboards were created to analyze inventory and fulfillment performance.

Dashboard 1: Inventory Overview
Shows product stock levels and warehouse distribution.

Dashboard 2: Order Fulfillment Analysis
Displays order processing time and delivery status.

Dashboard 3: Product Demand Trends
Shows most ordered products and sales patterns.

Important KPIs:

- Inventory Turnover Rate
- Order Processing Time
- Delivery Success Rate
- Stock Availability

These metrics help companies understand supply chain performance and identify bottlenecks.

4.3 AI Tool Integration Results

AI tools such as Chat GPT or Microsoft Copilot can assist in:

- Data analysis
- Report generation
- Insight extraction

Example prompts used:

- Identify trends in order fulfillment data
- Summarize inventory performance
- Suggest improvements for supply chain efficiency

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The analysis revealed several important insights:

1. Some products had very high demand but low inventory levels.
2. Certain warehouses had slower order processing times.
3. A small number of products contributed to the majority of orders.

These insights can help businesses optimize stock levels and improve delivery performance.

5.2 Limitations

Discuss the limitations of the current project or approach honestly.

1. Dataset used was limited in size
2. Real-time inventory tracking was not implemented
3. Analysis focused mainly on visualization

5.3 Future Work

Provide suggestions for improving or extending the project in the future.

1. Integration with real-time databases
2. Machine learning demand forecasting
3. Automated inventory alerts

4. Web-based dashboard systems

5.4 Conclusion

This project demonstrates how Excel and Power BI can be used to analyze inventory and order fulfillment data effectively.

By cleaning and analyzing data, businesses can gain insights into stock levels, order processing times, and product demand. Interactive dashboards provide clear visualizations that support better decision making.

The project successfully achieved its objectives and highlights the importance of data analytics in supply chain management.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

1. Microsoft Power BI Documentation
2. Microsoft Excel Documentation
3. Edunet Foundation AICW Fellowship Program
4. Supply Chain and Inventory Management Research Articles

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A – Sample Dataset
Appendix B – Excel Formulas Used

Examples:

COUNTIF
VLOOKUP
IFERROR

Appendix C – AI Prompts Used